CPM: FINDING CRITICAL PATH

Forty-seventh Lecture by Prof. Uday S. Dixit, Department of Mechanical Engineering, IIT Guwahati



Example

Activity	Predecessors	Duration (in days)
A	Nil	3
B	A	2
C	A	3
D	Nil	2
E	D	2
F	B, E	7
G	B, E	5
H	F, J	6
I	E, C	2
J	E, C	3

(a) Draw the activity network diagram.

(b) Determine the critical path and critical activities.

(c) Find the total project duration.

Activity on arrow (AOA)

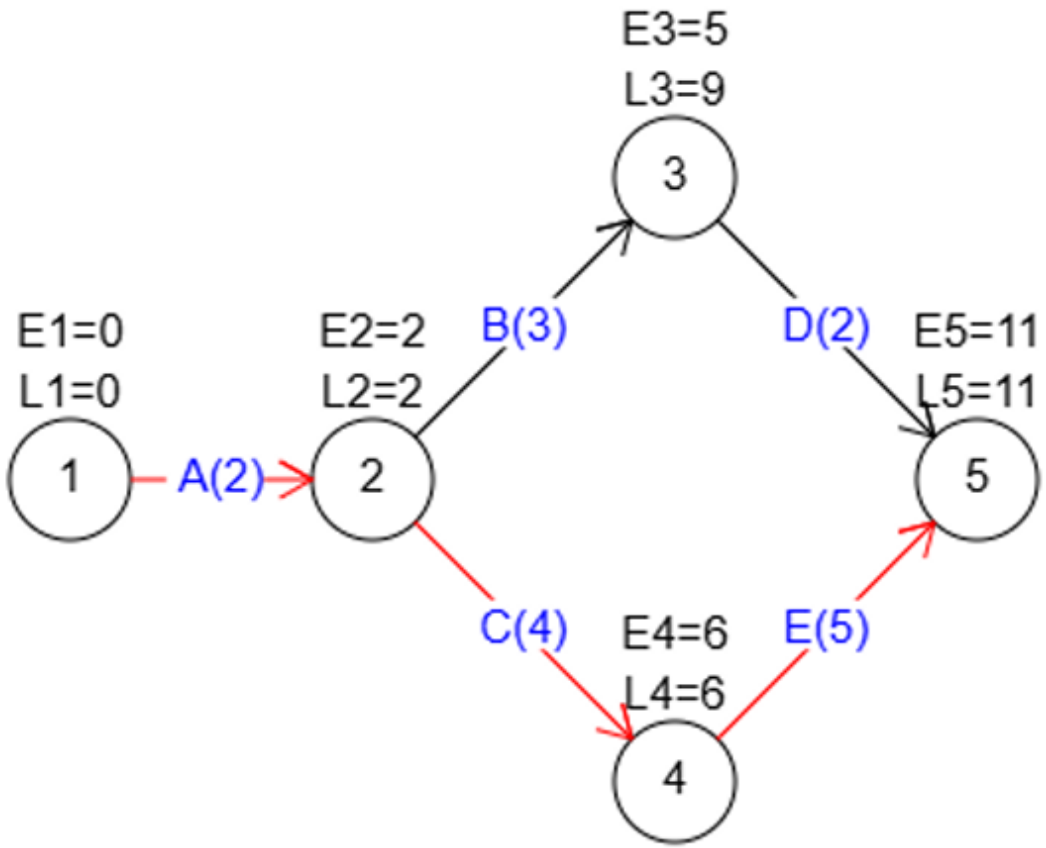


Figure: Network diagram

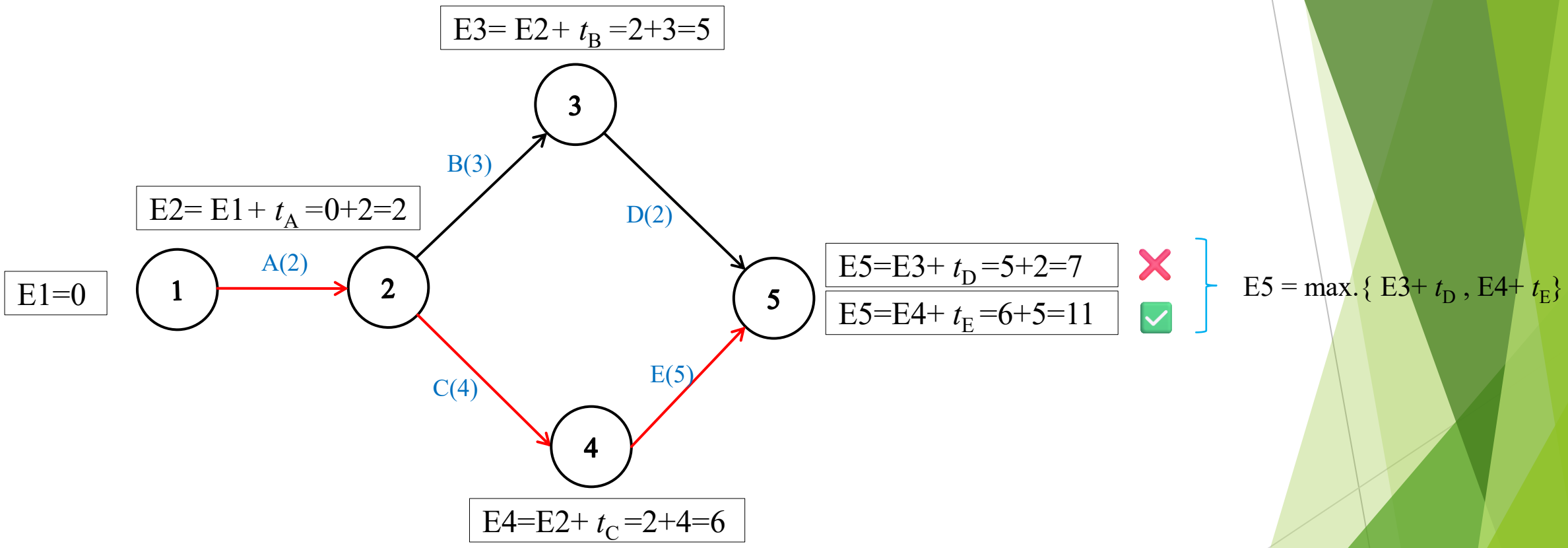
E: Earliest occurrence time of an event

L: Latest occurrence time of an event

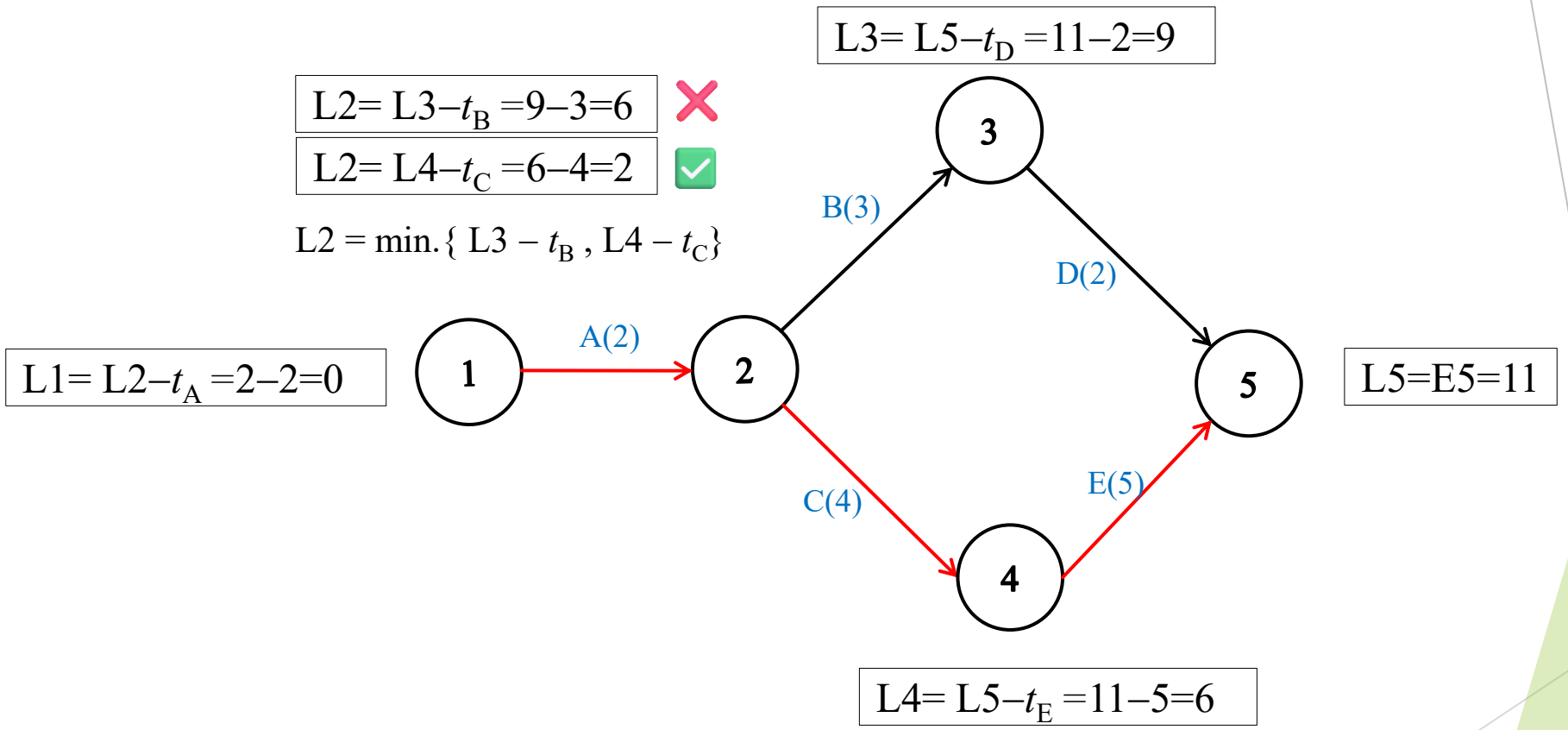
- Use forward pass to calculate earliest occurrence times. Start at **node 1** and advance to **node 5**.
- Set E = 0 to indicate that the project starts at time 0.

- Use backward pass to calculate latest occurrence times. Start at **node 5** and advance to **node 1**.
- Set L5 = E5 to indicate that the latest occurrence of the last node equals the project duration.

Forward pass



Backward pass



Critical path is shown in red colour.

The critical path of the project is: **1–2–4–5** and critical activities are **A**, **C**, and **E**.

Total project duration = sum of duration of critical activities

$= (2 + 4 + 5) = 11$ days.

Critical activity

- An activity (I,J) will be critical if it satisfies three conditions:

1. Latest occurrence time of event I= Earliest occurrence time of event I
2. Latest occurrence time of event J= Earliest occurrence time of event J
3. Earliest occurrence time of event J– Earliest occurrence time of event I

=Activity Duration = Latest occurrence time of event J– Latest occurrence time of event I

Critical activity

- When an activity's ES + duration = LF, or equivalently, when its float (LF − EF) is zero, it is a critical activity

Example

Activity	Predecessors	Duration (in days)
A	Nil	3
B	A	2
C	A	3
D	Nil	2
E	D	2
F	B, E	7
G	B, E	5
H	F, J	6
I	E, C	2
J	E, C	3

(a) Draw network diagram.

(b) Find out the critical path and critical activities.

(c) Find out the minimum time of project completion.

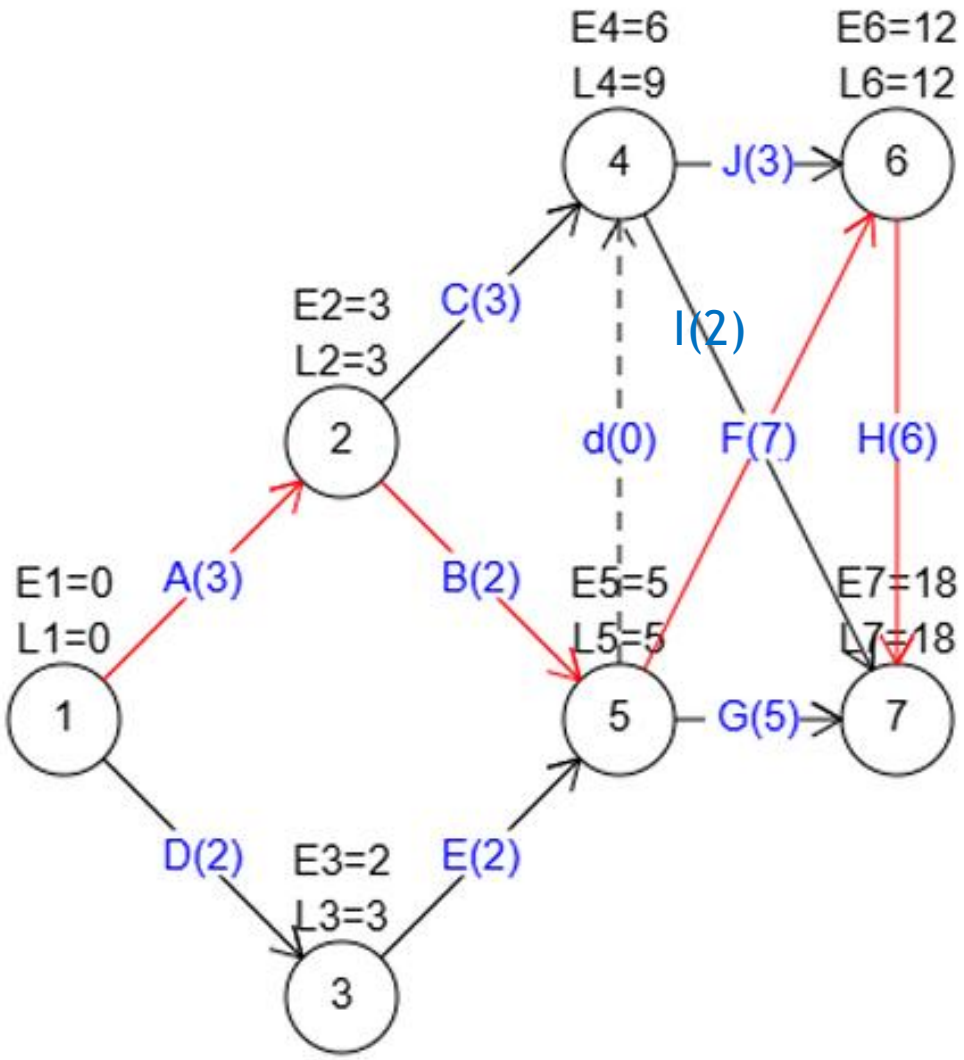
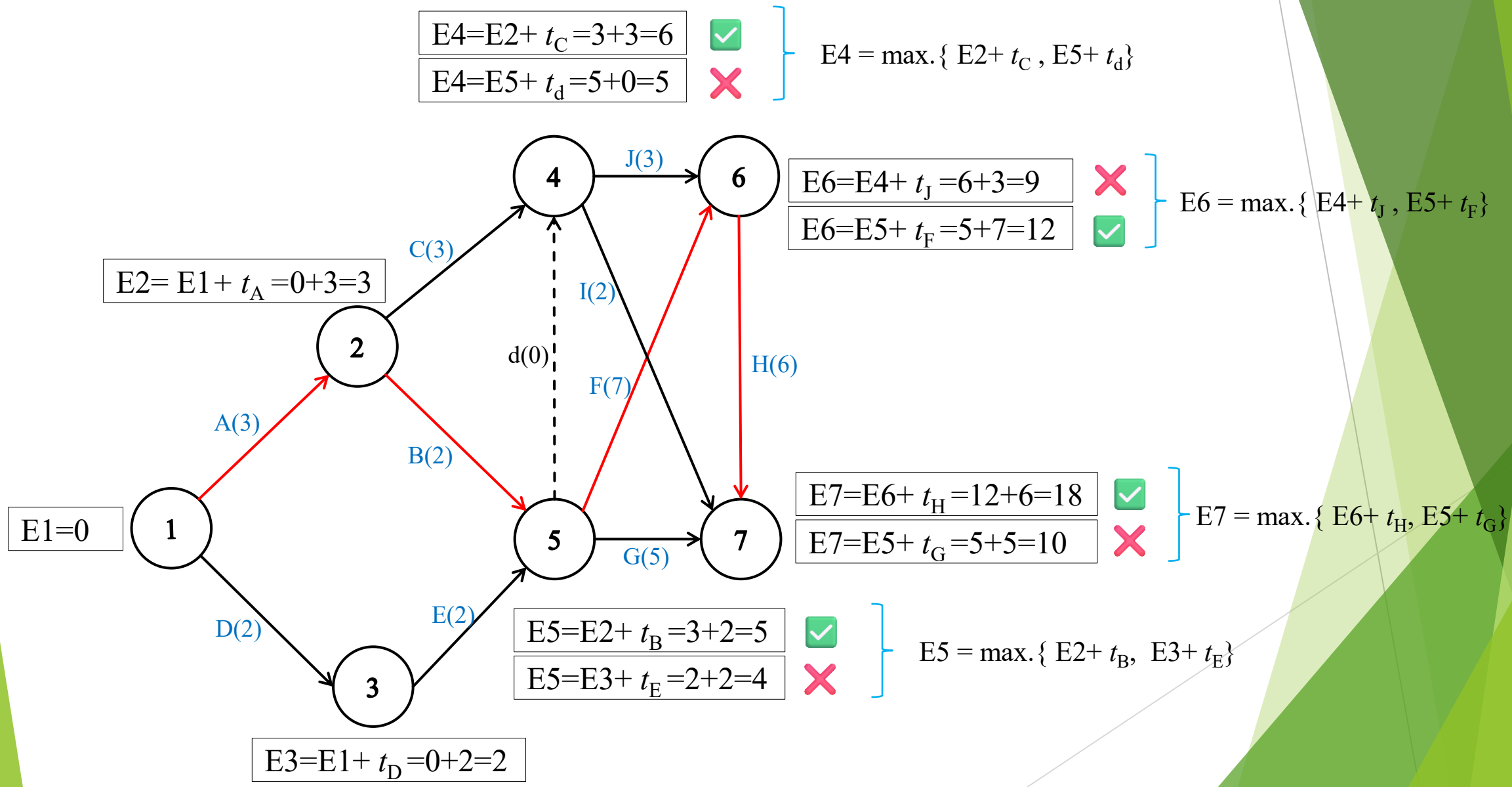


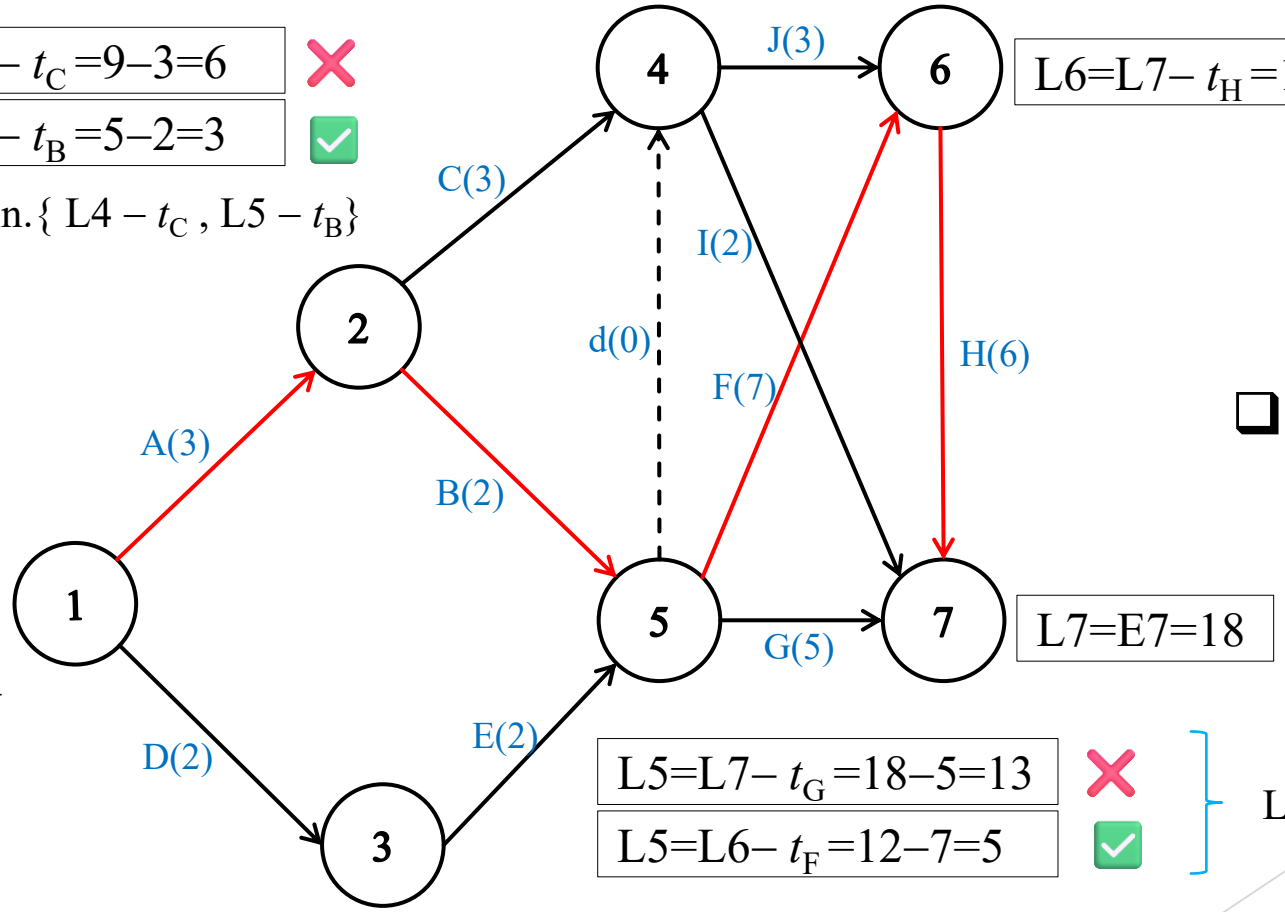
Figure: Network diagram

Observe how dummy activity is included in the network diagram

Forward pass



Backward pass



L4=L6– t_J=12–3=9 ✅

L4=L7– t_I=18–2=16 ❌

L4 = min.{ L6 – t_J , L7 – t_I}

L2=L4– t_C=9–3=6 ❌

L2=L5– t_B=5–2=3 ✅

L2 = min.{ L4 – t_C , L5 – t_B}

L6=L7– t_H=18–6=12

L1=L2–t_A=3–3=0 ✅

L1=L3– t_D=3–2=1 ❌

L1 = min.{ L2 – t_A , L3 – t_D}

L7=E7=18

L5=L7– t_G=18–5=13 ❌

L5=L6– t_F=12–7=5 ✅

L5 = min.{ L7 – t_G , L6 – t_F}

L3=L5– t_E=5–2=3

- Start from last node 7

Critical path is shown in red colour.

The critical path of the project is: **1–2–5–6–7** and critical activities are **A**, **B**, **F**, and **H**.

Minimum time of project completion = sum of duration of critical activities

= (3 + 2 + 7 + 6) = 18 days.

Example

Activities	Predecessor (s)	Duration , t (weeks)
A	--	3
B	--	2
C	A	4
D	A, B	2
E	C	4
F	C	3
G	D, E	6
H	F, G	2

Find the Earliest Start (ES), Earliest Finish (EF), Latest Start (LS), Latest Finish (LF) for each activity.

To find ES and EF, we use forward Pass

Activity A:

ES = 0

EF =0 + t_A = 0 + 3 = 3

Activity B:

ES = 0

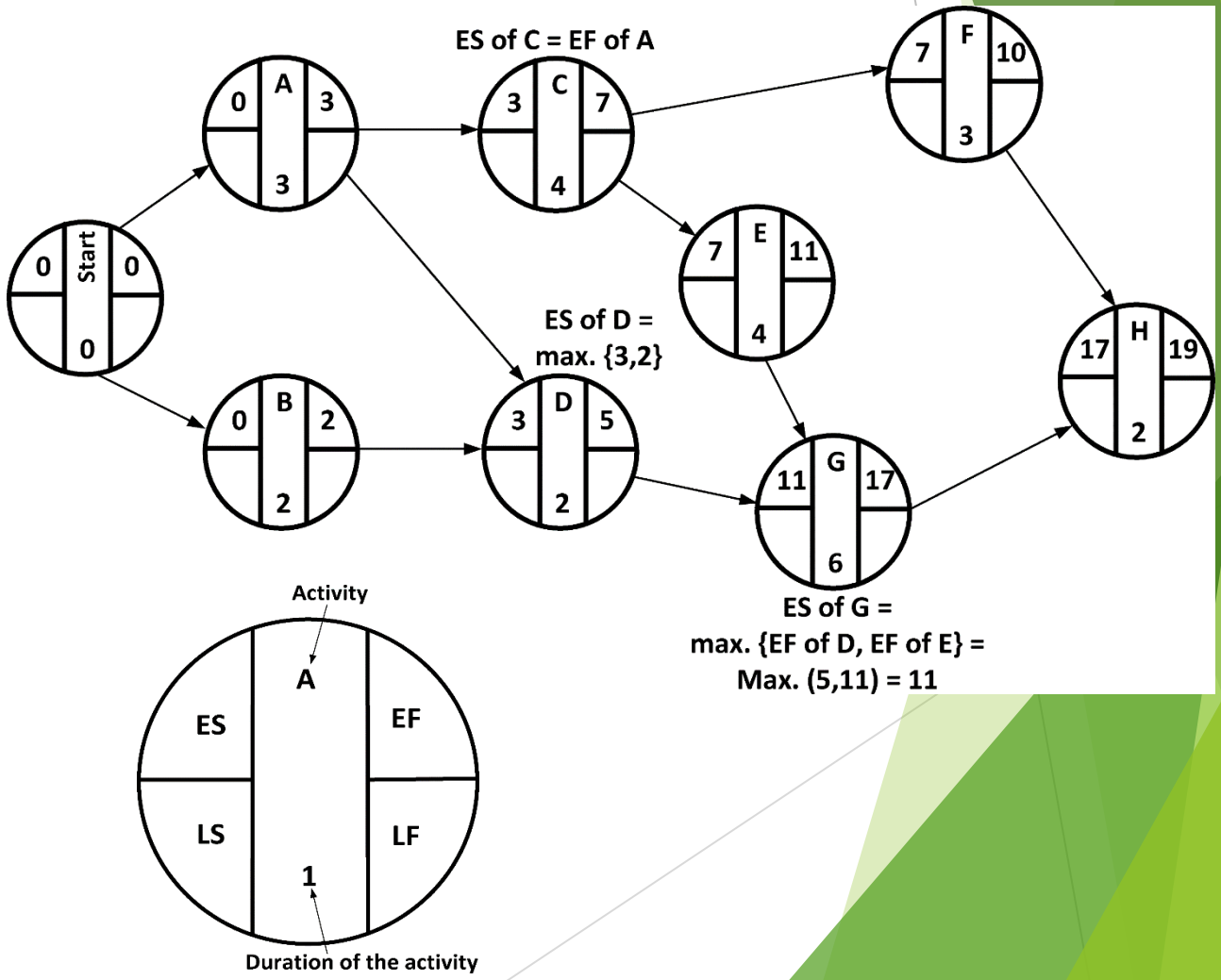
EF = 0 + t_B = 0 + 2 = 2

Activity C:

ES = EF of A = 3

EF = ES + t_C = 3 + 4 = 7

Activity on node (AON)



Activity D:

ES = Max (EF of A, EF of B) = Max (3, 2) = 3

EF = ES + t_D = 3 + 2 = 5

Activity E:

ES = EF of C = 7

EF = ES + t_E = 7 + 4 = 11

Activity F:

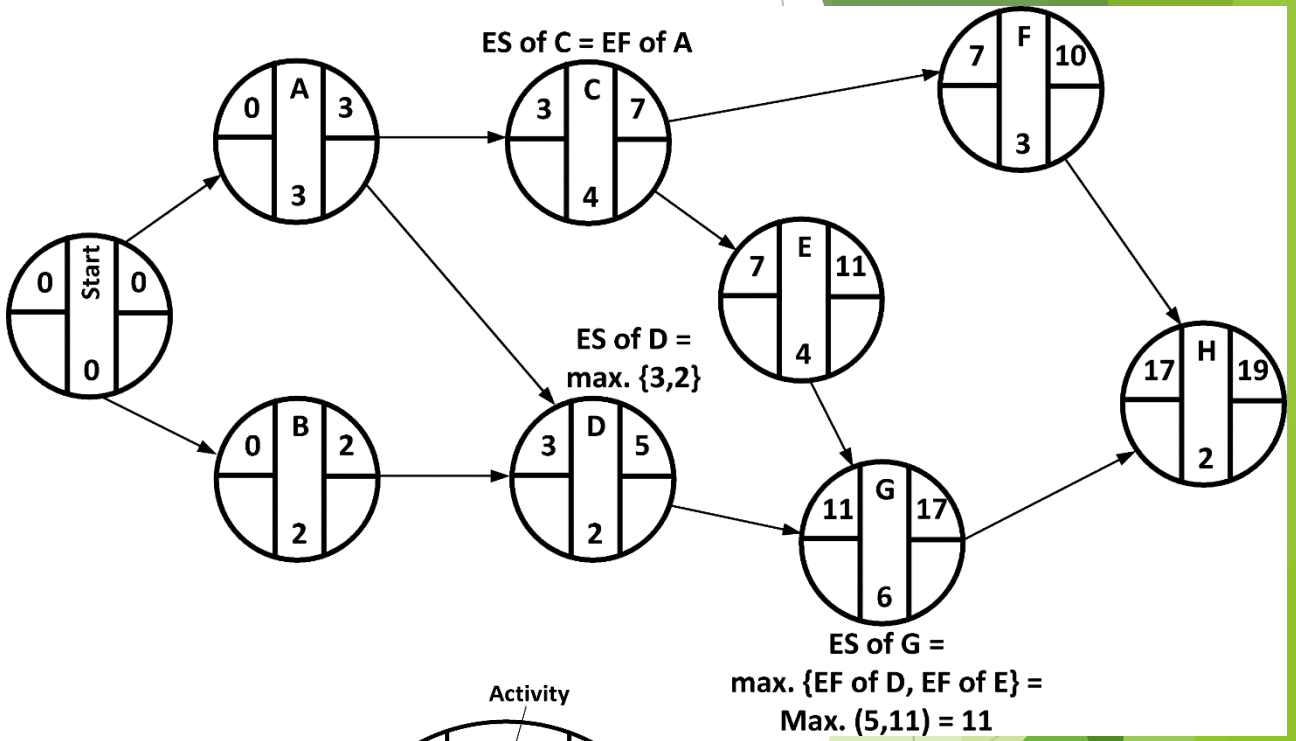
ES = EF of C = 7

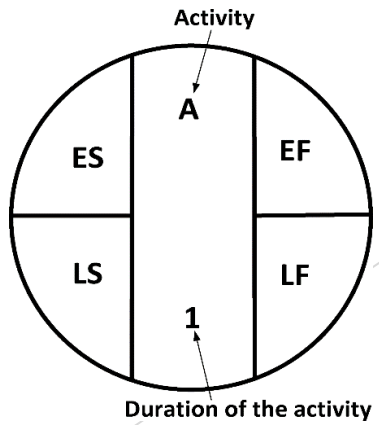
EF = ES + t_F = 7 + 3 = 10

Activity G:

ES = Max (EF of D, EF of E) = Max (5, 11) = 11

EF = ES + t_G = 11 + 6 = 17





Activity H:

ES = Max (EF of F, EF of G) = Max (10, 17) = 17

EF = ES + t_H = 17 + 2 = 19

- **To calculate LS and LF, we use backward pass.**

  Start from the last.

Activity H:

LF = 19

LS = LF – t_H = 19 – 2 = 17

Activity G:

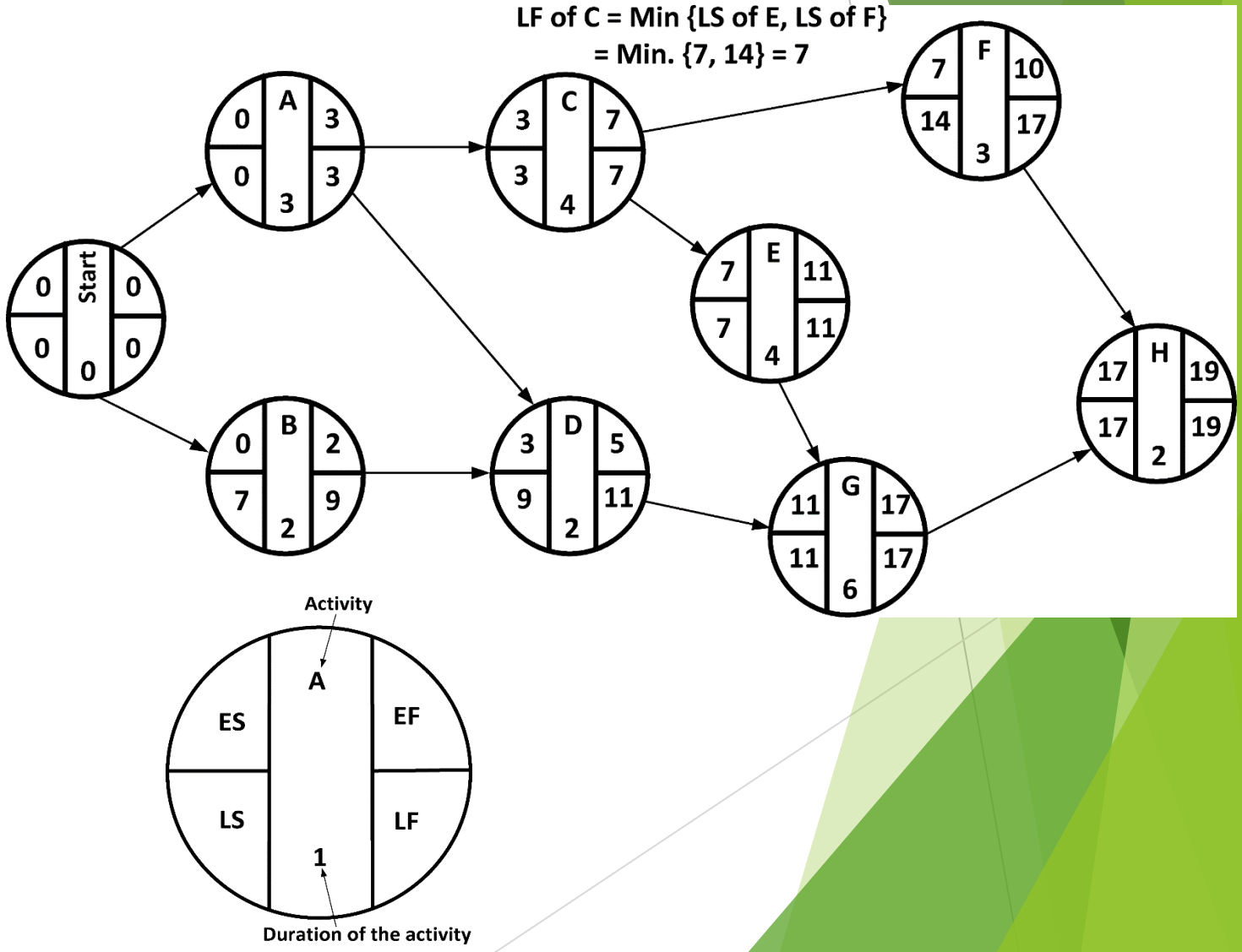
LF = LS of H = 17

LS = LF – t_G = 17 – 6 = 11

Activity F:

LF = LS of H = 17

LS = LF – t_F = 17 – 3 = 14



Activity E:

LF = LS of G = 11

LS = LF – t_E = 11 – 4 = 7

Activity D:

LF = LS of G = 11

LS = LF – t_D = 11 – 2 = 9

Activity C:

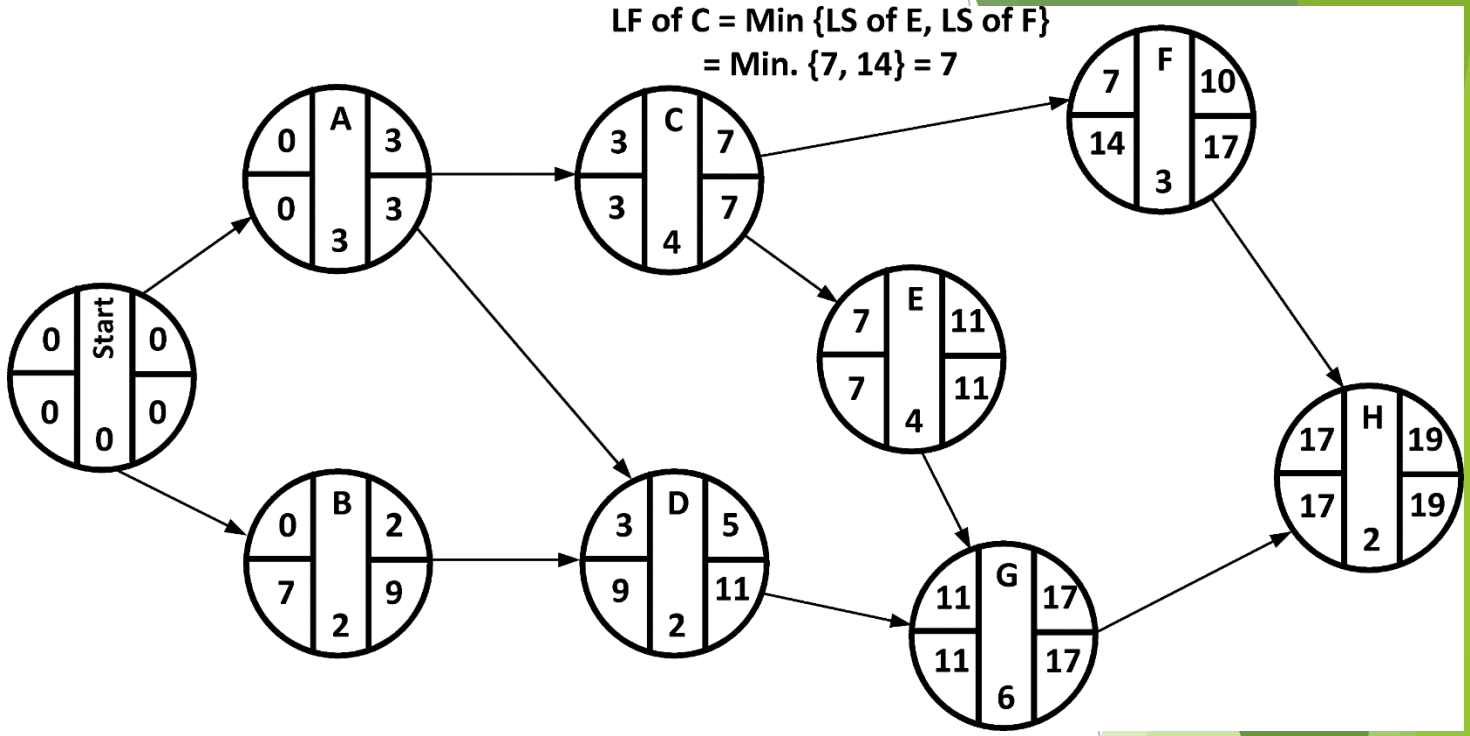
LF = Min (LS of E, LS of F) = Min (7, 14) = 7

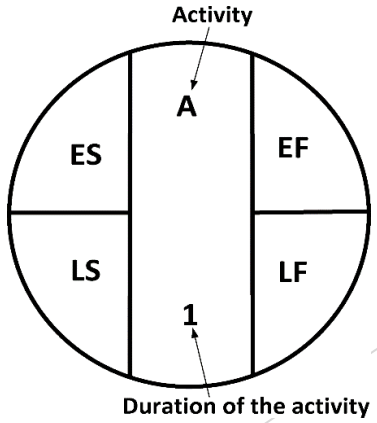
LS = LF – t_C = 7 – 4 = 3

Activity B:

LF = LS of D = 9

LS = LF – t_B = 9 – 2 = 7





Activity A:

LF = Min (LS of C, LS of D) = Min (3, 9) = 3

LS = LF – t_A = 3 – 3 = 0